

Modular L1-to-RL Manipulation Stack with Route-Curriculum Extension

****Course:**** ENPM690 Final Project

****Project:**** Robot Brain Trainer

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1. Abstract

This project implements a modular L1/L2/L3 robot learning stack for kitchen-style manipulation. L1 is a Qwen semantic bridge that turns a natural-language command into a validated `IntentPacket`; L2/L3 contain the learned kinematic manipulation stack and deterministic execution boundary. The Phase 2 controller was simplified to `Approach -> Finisher`, which reaches 93% success in the hardest current kinematic workspace stage with approximately 2.89 mm final position error and 0.0208 rad final orientation error. A route-curriculum extension further expands the learned controller from local waypoint correction to long-prefix route following: the original full-route probe had longest prefix 21, while the route-trained policy reaches prefix120 with 100% sequential success and reaches prefix170 in a full 483-waypoint probe. Full holder1-to-holder8 Gazebo kitchen transport is not yet solved.

2. Motivation

Kitchen manipulation is difficult to debug as a monolithic end-to-end system. Semantic interpretation, learned motor control, and low-level execution have different failure modes. This project therefore separates the stack into three layers:

- L1: semantic intent using Qwen / VLM-style reasoning.
- L2: learned RL skill policy.
- L3: deterministic execution and safety boundary.

This decomposition makes the system easier to inspect: the LLM cannot directly move the robot, and the motor policy does not need to solve natural-language understanding.

3. Original Proposal vs Final Scope

The original proposal targeted a modular kitchen manipulation pipeline:

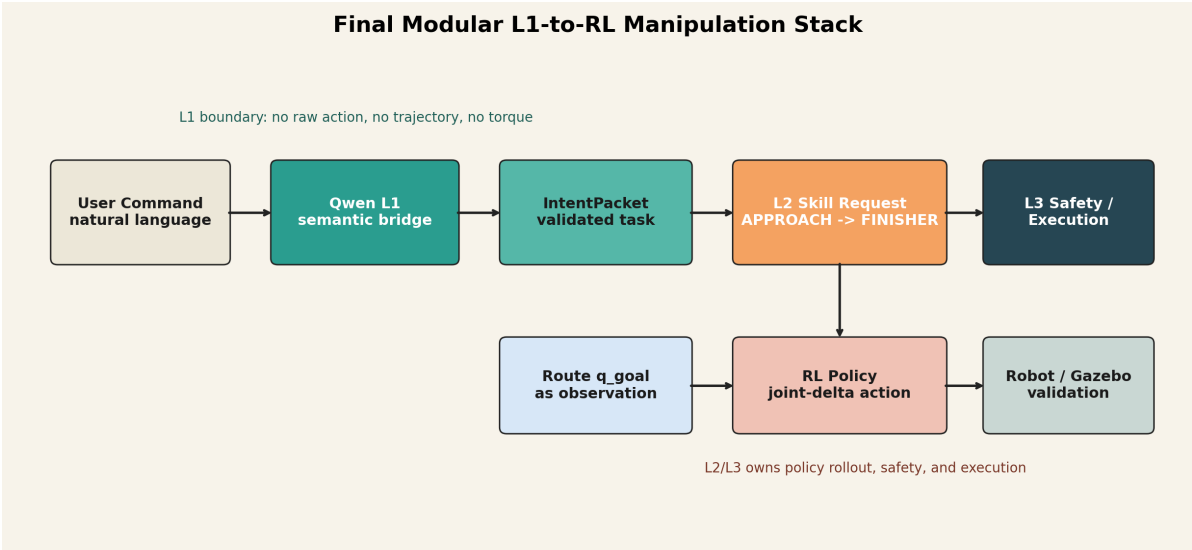
L1 semantic interpretation -> L2 learned skill -> L3 safe execution

The final scope is a structured proof-of-concept:

- A Qwen bridge that produces validated structured commands.
- A kinematic RL skill stack.
- Route-curriculum numeric validation.

The final system does not claim full Gazebo physics completion, real camera grounding, or real robot deployment.

4. System Architecture



The key contract is that L1 never emits raw controls. Qwen can choose object, source slot, target slot, and constraints. L2/L3 own policy inference, safety, and execution.

5. Phase 1 Baseline

Phase 1 established the research infrastructure:

- Gymnasium kinematic environment.
- PPO/TD3 training.
- Deterministic evaluation.
- Isolated approach and dock policies.

However, Phase 1 did not solve final integration. Approach/dock switching was unreliable, and the VLM/Qwen component had not started. Phase 1 proved infrastructure rather than final behavior.

6. Phase 2 Skill Stack

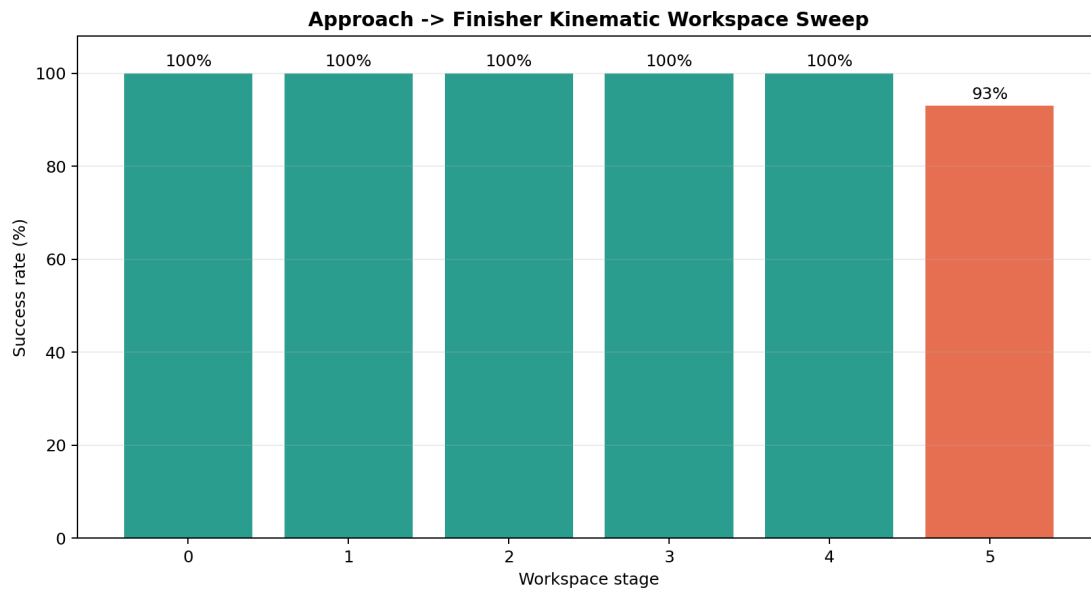
Phase 2 simplified the final control path to:

Approach -> Finisher

Dock-Coarse, Bridge, readiness classifier, acceptance maps, and finisher adaptation were diagnostic tools. They helped reveal that adding modules was not always helpful; once Approach could produce clean handoff states, the cleanest final path was `Approach -> Finisher`.

Stage	Success	Handoff Pos Error	Handoff Ori Error	Final Pos Error	Final Ori Error
---	---	---	---	---	---
0	1.00	0.50 mm	0.0073 rad	1.67 mm	0.0106 rad
1	1.00	0.62 mm	0.0099 rad	1.67 mm	0.0123 rad
2	1.00	0.85 mm	0.0119 rad	1.82 mm	0.0139 rad
3	1.00	1.20 mm	0.0138 rad	2.14 mm	0.0164 rad
4	1.00	1.71 mm	0.0150 rad	2.53 mm	0.0165 rad
5	0.93	1.96 mm	0.0177 rad	2.89 mm	0.0208 rad

The most demanding Stage 5 reached 93% success, with about 2.89 mm final position error and 0.0208 rad final orientation error.



7. Qwen L1 Bridge

The Qwen bridge exposes a structured tool interface:

- ``get_l1_scene_context``
- ``resolve_intent_packet``
- ``prepare_phase1_skill_request``

For the command:

Move tray1 from shelf_A1 to shelf_B1 while keeping it level and inserting with a stable pose.

Qwen produces a structured call to ``resolve_intent_packet``, resolving:

```
object_id: tray1<br/>source_slot: shelf_A1<br/>target_slot: shelf_B1<br/>constraints: speed_cap = SLOW<br/>resolved command: MOVE_PLATE(shelf_A1, shelf_B1)<br/>pipeline: APPROACH -> FINISHER<br/>target pose: [-0.92, -1.16, 1.22]<br/>orientation: [3.14, 0.0, 3.14]
```

This validates the semantic-to-RL contract without letting the LLM emit low-level robot controls.

8. Route Curriculum Extension

The route curriculum was added after discovering that the policy was a strong local controller but not a full scene-level transport controller.

Before route curriculum:

full483 success rate: 0.0435
longest continuous success prefix: 21

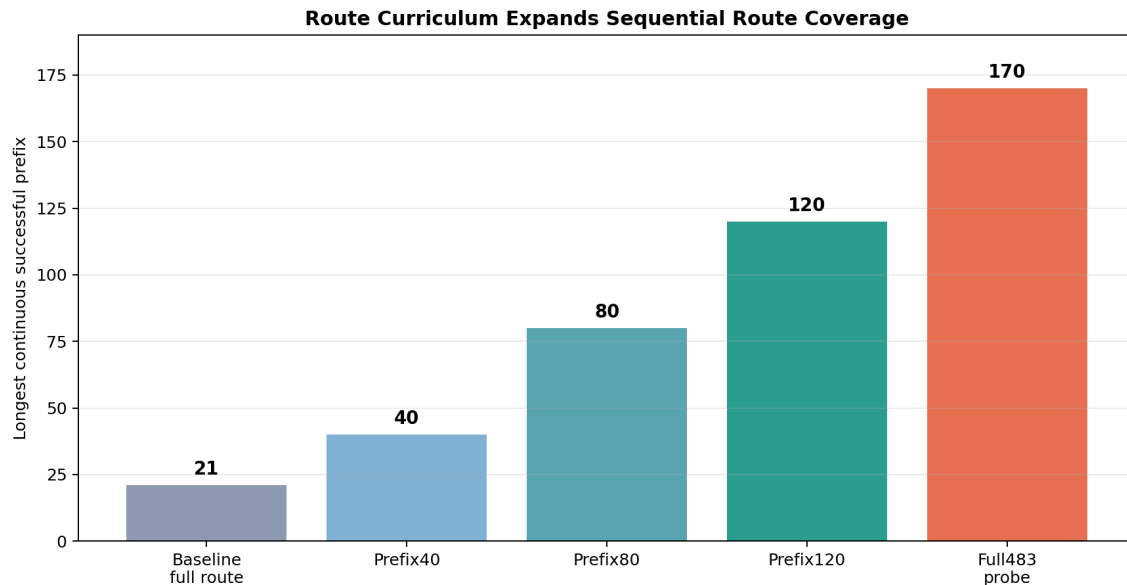
Implementation changes:

- Dense q-goal route target.

- Route-specific observation keys: `route\_q\_goal`, `route\_q\_error`, `route\_tangent`, `route\_scalar`.
- Prefix curriculum.
- Sequential actual-final-q evaluation.

After route curriculum:

prefix120 sequential success: 1.0
 prefix120 route distance: 1.720 m
 full483 probe success: 0.4741
 full483 longest prefix: 170
 first failure index: 171
 first failure reason: position

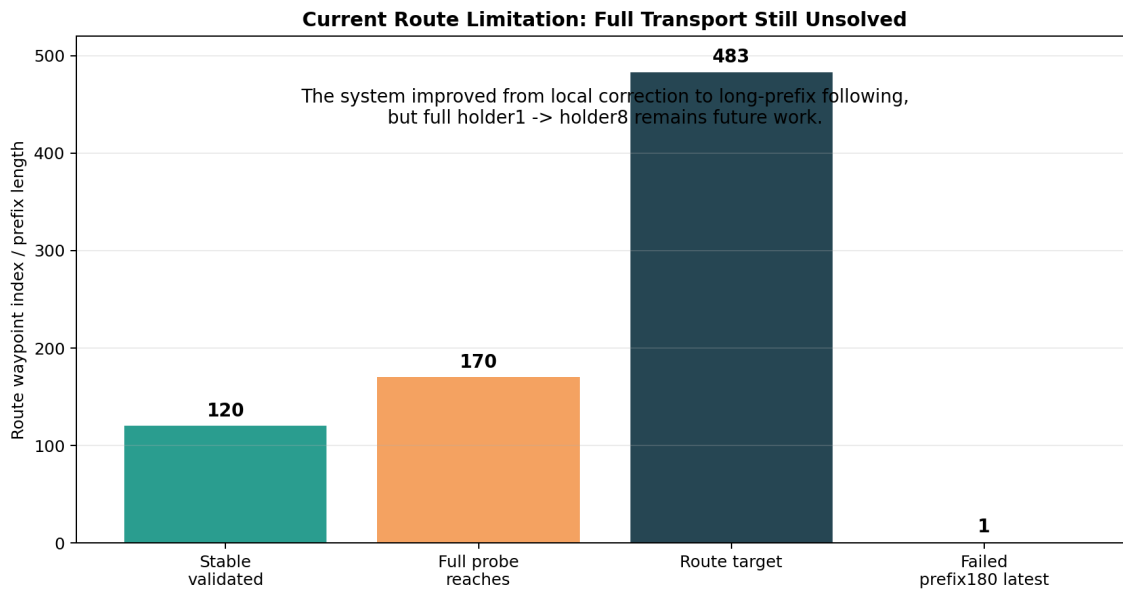


9. Negative Results and Lessons

Several negative results are central to the final interpretation:

- Prefix180 direct fine-tune failed early-prefix retention.
- Prefix180 anti-forgetting retry also failed.
- Sampled training success can be misleading.
- Sequential actual-final-q evaluation is the real route metric.
- Teacher-anchor smoke was promising, but a longer latest checkpoint drifted.

This means future training needs sequential-gated checkpoint selection, not only longer PPO fine-tuning.



10. Phase 3A Real Gazebo Evidence

The final package now includes a real Gazebo / ROS2 controlled-sim evidence run, separate from the report-style videos.

Verified artifact:

video: report/videos/real_gz_camera_phase3a_controlled_sim.mp4
 run id: final_real_gz_sensor_demo_005
 step log: artifacts/v5/phase3a_controlled_sim/final_real_gz_sensor_demo_005/runtime_steps.jsonl

This run launched the kitchen scene, recorded the Gazebo side camera topic `/v5/cam/side/rgb`, loaded the frozen `Approach -> Finisher` policies, and sent commands through the ROS action path `/arm_controller/follow_joint_trajectory`.

Metric	Value
Target count	3
Runtime steps	60
Action goals executed successfully	60 / 60
Success rate on visible-workspace FK targets	0.3333
Mean final position error	0.04586 m
Mean final orientation error	0.03366 rad
Mean tracking error L2	0.0000807
Gazebo camera frames captured	251

This result should be interpreted carefully. It proves that the sim integration path is real and that the arm is commanded through the Gazebo/ROS controller stack, but it does not solve the complete holder1 -> holder8 tray transport. The best quantitative success result remains the kinematic `Approach -> Finisher` workspace sweep, while this Phase 3A run provides honest simulation evidence and exposes the remaining sim-transfer gap.

11. Limitations

The system does not yet solve:

- Full holder1 -> holder8 route transport.
- Full Gazebo physics validation.
- Image-grounded Qwen-VL perception.
- Contact, friction, and tray dynamics.
- Real robot deployment.

12. Future Work

Route Curriculum

- Sequential-gated checkpoint selection.
- Teacher-anchor early stopping.
- Prefix180 / prefix260 expansion.

Gazebo Validation

- Move the route-trained checkpoint into Phase 3A runtime.
- Validate controller timing and joint execution.
- Add collision/contact/object stability checks.

Perception

- Replace structured scene proxy with image-grounded Qwen-VL scene estimation.
- Project paper/object geometry into a ground-plane coordinate frame before using visual headings.

13. Conclusion

This project demonstrates a modular architecture, a working kinematic `Approach -> Finisher` skill stack, a Qwen semantic bridge, and a route-curriculum extension from local correction to long-prefix route following. Full kitchen transport remains future work, but the modular learning stack and route-curriculum direction are validated by quantitative results.